Dr. Rathin K. Joshi

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Research Profile: Postdoctoral researcher at Johns Hopkins University specializing in neural signal processing, EEG/ERP analysis, brain-computer interfaces, and machine learning. Research spans computational neural biomarkers, interpretable modeling, wearable neurotechnology, and translational applications.

EDUCATION

- **Ph.D., Electronic Systems Engineering**
  IISc Bangalore, 2018–2024 | CGPA: 7.9/10
  Thesis: Event-Related Potential Interpretation Approaches for Neonatal Hearing Screening [Link]
  Submitted: Dec 2023, Defended: Apr 2024.
- **M.Tech., VLSI and Embedded Systems**
  DA-IICT, Gandhinagar, 2013–2015 | CGPA: 7.55/10
  Thesis: SET-based 4-bit ALU Design, Simulation & Optimization [Link]
  Submitted: May 2015, Defended: Jul 2015.

PROFESSIONAL EXPERIENCE

- **Postdoctoral Fellow, Johns Hopkins University**
  Baltimore, USA — Oct 2025 – Present
  - Develop computational EEG analysis methods for neural research.
  - Apply machine learning, signal processing, and control-theoretic approaches to neural data.
  - Mentor doctoral and master's-level researchers.
- **Postdoctoral Research Associate, IISc**
  Bangalore, India — Apr 2024 – Apr 2025
  - Supported technology transfer of doctoral neurotechnology research
  - Supported neural signal analysis in invasive rodent electrophysiology models.
  - Mentored postgraduate and doctoral researchers.
- **Systems Engineer, TCS**
  Pune, India — Aug 2015 – Dec 2017
  - Developed and validated modular automotive control logic for Ford Motor Company.
  - Supported MATLAB-based model development and coordination with testing teams.

KEY RESEARCH PROJECTS

- **AI/ML based ADHD classficiation**: Interpretable spectral and dynamical biomarkers for ADHD classification.
- **Neonatal Hearing Screener**: Automated extraction of cortical and brainstem auditory responses
- **Epileptic Seizure Detection and Classification**: Mimics visual inspection of interictal EEGs for epilepsy classification.
- **Cognitive Screener**: Extracts visually evoked responses for attention and memory screening.
- **Neural Signal Interpretation**: Analyzes ECoG/sEEG data for epilepsy and Parkinson's using rodent MEAs.

SKILLS

Expertise	Neural Signal Processing, EEG/ERP, Brain-Computer Interfaces, Computational Neural Biomarkers, Neurotechnology
Methods	ERP Extraction, Dynamic Network Modeling, Spectral Analysis, Machine Learning, Statistical Analysis, Feature Engineering, Model Validation
Coding	Python, MATLAB, Scientific Computing, Debugging, Automation, Git/GitHub
Tools	MNE-Python, EEGLAB, ERPLAB, Origin, COMSOL, SolidWorks, LaTeX
Practical	EEG/ERP Acquisition, Biomedical Instrumentation, Wearable Systems, Electrophysiology, Prototyping, Technology Transfer
Ethics	Ethical Protocols, Regulatory Awareness, Research Documentation

SELECTED PATENTS TOTAL: 6+

- **Neonatal Headgear Interface** –Granted: Jun 2026
- **L-shaped Stimulation Implant** –Granted: Sep 2025
- **EEG Seizure Detector & Classifier** –Granted: Jan 2025
- **Sensory Pathway Scanner** –Granted: Apr 2024

SELECTED PUBLICATIONS TOTAL:13+

- Joshi et al., "Automated ABR and MMN using customized headband for Hearing Screening," *Biomed Sig Proc & Cont*, 2024. [Link]
- Joshi et al., "Spatiotemporal EEG analysis for seizure detection," *Biomed Sig Proc & Cont*, 2023. [Link]
- Chatterjee et al., "Microelectrode Array for Rodent Epileptic Models," *Biomed Microdevices*, 2023. [Link]
- Joshi et al., "P300 ERP Extraction using Wearable Device," *IEEE BIOCAS*, 2022. [Link]

AWARDS AND HONORS

- SITARE-GYTI Award for Neonatal Screener [Link]
- SHIFT Health Hackathon– 1st Runner-up [Link]
- SmartX Health Hackathon – 1st Runner-up[Link]

SELECTED DEMONSTRATIONS TOTAL: 20+

- Towards Single-Event P300 Extraction.
  Project Completion Demonstration, (Jun. 08-09, 2022)
  DIPAS, DRDO, Delhi [Link]
- ERP Extractor Development and Applications.
  BCL Workshop, (Jan. 09-13, 2023)
  Faculty Hall, Indian Institute of Science, Bangalore. [Link]

TEACHING

- **Neuroscience & Neuroinstrumentation** *(Online)*:
  [BCI Basics], [EEG Processing], [Microelectrode Arrays]
- **Advanced Neural Science for Engineers** *(Online)*:
  [Lithography], [Fourier Optics], [EEG Applications]
- **Mathematical Aspects of BioSystems** *(Online)*:
  [Signal Classification], [Signal Acquisition]
- **Analog Circuit Design** *(Offline)*:
  Topics Taught: OpAmp Basics, Arithmetic Circuits, ADC, DAC, Clippers, Clampers
- **Engineering Mathematics** *(Offline)*:
  Topics Taught: Calculus, Linear Algebra

ADDITIONAL ACHIEVEMENTS

- Intercollegiate chess player (UG & PG).
- MATLAB Cody: World Rank 351 / 148,038.
- GATE 2013: AIR 1667 / 2,56,135 (ECE).
- CSIR NET 2012: AIR 40 / 27216 (Engineering Sciences).